

Juan Lopez Rios

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Profile

Data Scientist with a strong physics, machine learning, and data analysis foundation. Experienced in research-based problem-solving, data engineering, and deploying machine learning models. Experienced in Python, machine learning libraries, SQL, and competent in AWS. Skilled at translating complex technical concepts into actionable insights. Seeking to contribute to innovative projects in high-performance data-driven environments.

Skills & Languages

Programming & Scripting: Python (NumPy, Pandas, Scikit-learn, TensorFlow, PyTorch, Matplotlib), **SQL** (SQLite).
Machine Learning & AI: Supervised/Unsupervised Learning, Deep Learning, NLP, Computer Vision, MLOps workflows.
Data engineering & Deployment: MLflow, Mage, Docker, Streamlit, Solara, Langchain, AWS, API (Flask).
Mathematical & Scientific Tools: Wolfram Mathematica, GROMACS, PyMol.
Languages: English (Native), Spanish (Native), Japanese (Basic)

Professional Experience

Computational Researcher, ([UCL](#)) **London, UK** 09/2023 - 06/2024

- **Machine Learning Model Development:** Implemented machine learning models (Random Forest, Neural Networks, Gradient Boosting, Support Vector Machines, and TabNet) to identify optimal ligands from a dataset of 10,000+ candidates targeting oncogenic Ras proteins. Achieved a 0.5 improvement in Pearson correlation, enhancing predictive accuracy significantly.
- **Data Pipeline Engineering:** Engineered scalable data pipelines with 1,018+ features using RDKit, PyMOL, and equilibrium simulations, constructing new datasets and improving target ligand identification.
- **Academic Excellence:** Conducted thesis research on predictive modeling of oncogenic Ras-mutated protein reactivation, earning first-class honors (80%) and demonstrating advanced expertise in modeling techniques and computational biophysics.

Machine Learning Intern, ([ENCORD](#)) **London, UK** 05/2023 - 08/2023

- **Enhanced SAM for Segmentation:** Fine-tuned and analyzed the Segment Anything Model (SAM) to optimize efficiency, reduce computational cost, and improve segmentation accuracy for over 20 specific objects. Integrated mean average masks and advanced pre-processing techniques to enhance functionality within Encord's AI annotation platform.
- **Comparative Analysis of Segmentation Models:** Benchmarked Per-SAM-f against Mask R-CNN on the DeepFashion dataset, achieving a mean Average Precision (mAP) of 0.501, surpassing Mask R-CNN in performance. The findings were published in a blog post on Encord's website, showcasing expertise in model evaluation and dataset-specific insights.
- **Developed Python-Based AI Chatbot:** Built an AI chatbot using the OpenAI API and LangChain framework for product navigation and customer support. Automated product data collection with a Python web scraper, boosting query efficiency and enabling personalized responses for Encord's offerings.

Research Intern, ([KCL](#)) **London, UK** 05/2021 - 06/2021

- **Molecular Dynamics Simulations & Analysis:** Conducted MD simulations with GROMACS to analyze biomolecule-solvent interactions. Utilized Python for data analysis, advancing expertise in computational biophysics.

Education

MSc Theoretical Physics (First Class Honours) [University College London](#) **London, UK** 2023-2024

Relevant Courses: High-Performance Computing, Quantum Computing, Molecular Physics and Computational Biophysics.
Thesis: Machine-Learning Tools for Biocatalysis.

BSc Theoretical Physics (First Class Honours) [University College London](#) **London, UK** 2020-2023

Relevant Courses: Machine Learning for Physicist, Computational Mathematics, Mathematics for Physics & Astronomy and Theory of Dynamical Systems.

Thesis: Machine-Learning Based Ocular Imaging. **2nd Prize UCL Physics Department**

Certifications & Hackathons

- **Courses:** Data Structures & Algorithms in Python, SQL for Data Science by UC Davis, MLOps Zoomcamp
- **Optiver Hackathon:** Lead a team to develop a trading algorithm using Python and Machine Learning (Final Round)